

Experiment 12

12.1.1 Phone Number Database Management

Algorithm :

Step 1: Start

Step 2: Read n (number of commands)

Step 3: Initialize ph as an empty dictionary {}

Step 4: Set i = 0

Step 5: If $i < n$, go to Step 6. Else go to Step 11.

Step 6: Read command (cmd)

Step 7: If $\text{cmd}[0] == \text{"ADD"}:$
 Store $\text{ph}[\text{cmd}[1]] = \text{cmd}[2]$
 Go to Step 10

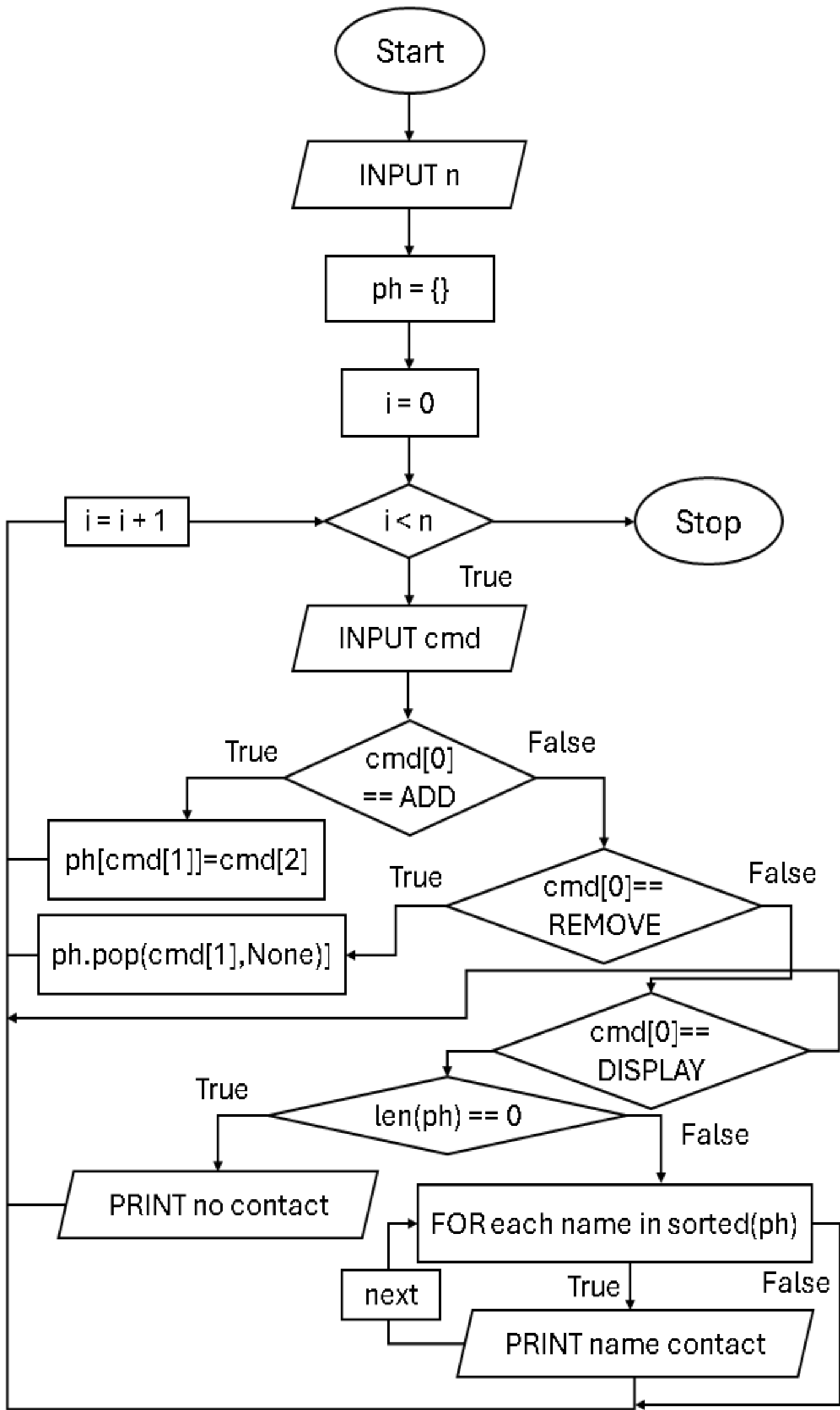
Step 8: Else if $\text{cmd}[0] == \text{"REMOVE"}:$
 Remove $\text{cmd}[1]$ from ph (if exists)
 Go to Step 10

Step 9: Else if $\text{cmd}[0] == \text{"DISPLAY"}:$
 If ph is empty:
 Print "No contacts"
 Else:
 For each name in $\text{sorted}(\text{ph}):$
 Print name and $\text{ph}[\text{name}]$
 Go to Step 10

Step 10: Set $i = i + 1$
 Go to Step 5

Step 11: Stop

Flowchart :



```
n = int(input())
ph = {}
for i in range(n):
    cmd = input().split()

    if cmd[0] == "ADD":
        ph[cmd[1]] = cmd[2]
    elif cmd[0] == "REMOVE":
        ph.pop(cmd[1], None)
    elif cmd[0] == "DISPLAY":
        if len(ph) == 0:
            print("No contacts")
        else:
            for name in sorted(ph):
                print(f"{name}: {ph[name]}")
```

Execution :

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12.1.1. Phone Number Database Management05:45

Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.

The program should support three operations: adding a contact (if the name already exists, update the phone number), removing a contact (if the name does not exist, ignore the operation), and displaying all contacts in alphabetical order by name.

Input Format:
1. First line contains an integer *n* representing the number of operations.
2. Each of the next *n* lines contains a single operation in one of the following formats:
• "ADD <name> <phone_number>" where <name> is a string, <phone_number> is a string of 10 digits.
• "REMOVE <name>" - name is a string.
• "DISPLAY" - no additional parameters.

Output Format:
• For display operation, print all contacts in alphabetical order by name in the format:
<name>: <phone_number>

• If the database is empty during display operation, print: "No contacts".
• Do not print anything for add and remove operations.

Constraints:
• Name contains only alphabetic characters (no spaces). And it is case sensitive.

Sample Test Cases

phoneDa...

1n = int(input())
2ph = {}
3
4for i in range(n):
5 cmd = input().split()
6
7 if cmd[0] == "ADD":
8 ph[cmd[1]] = cmd[2]
9
10 elif cmd[0] == "REMOVE":
11 ph.pop(cmd[1], None)
12
13 elif cmd[0] == "DISPLAY":
14 if len(ph) == 0:
15 print("No contacts")
16 else:
17 for name in sorted(ph):
18 print(f"{name}: {ph[name]}")

Average time0.022 s22.50 msMaximum time0.030 s30.00 ms5 out of 5 shown test case(s) passed3 out of 3 hidden test case(s) passed

Test case 124 msDebugTest cases

Expected output
4
ADD Alice 9876543218
ADD Bob 8765432189
ADD Charlie 777777777
DISPLAY
Alice: 9876543218

TerminalTest cases

< PrevResetSubmitNext>